Why Have House Prices in Turkey Gone Up in the Last Ten Years?

Project Final Report
EMU660 Statistical Learning and Analytics

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Abstract

Over the past decade, buying a home in Türkiye has become increasingly out of reach for ordinary households. This study asks what drove the sharp rise in housing prices between 2014 and 2024, using four official monthly time-series from the Central Bank of Türkiye (TCMB/EVDS): the House Price Index (HPI), housing sales, USD/EUR exchange rates, and credit card sectoral expenditure. Through exploratory analysis, correlation testing, regional comparison, and log-log regression, we find that currency depreciation and broad inflation are the dominant forces at work, while sales volumes actually move in the opposite direction to prices. Regional differences are large, yet our model still explains roughly 95–99% of the variance in the national HPI.

1 Introduction

Few economic issues have touched Turkish households more directly than the cost of housing. After years of moderate growth, the national House Price Index (HPI) took off around 2019–2020 and has not looked back since, leaving many families priced out of the market. This report asks:

Why have house prices in Turkey gone up in the last ten years?

To answer it, we draw on four monthly time-series from the Electronic Data Delivery System (EVDS) of the Central Bank of Türkiye (TCMB): the HPI, housing sales, USD/TRY and EUR/TRY exchange rates, and credit card sectoral expenditure. Section 2 describes the data, Section 3 walks through the analysis, and Section 4 pulls together the key findings.

2 Data

2.1 Data Source

Everything in this project comes from the **Electronic Data Delivery System (EVDS)** of the Central Bank of Türkiye (TCMB) — the standard repository for official, high-frequency economic data in Türkiye. We use four datasets:

- **House Price Index (HPI)** — monthly index at national and regional (NUTS2) levels.
- **House Sales Statistics** — monthly provincial sales data.
- **Exchange Rates (USD/TRY, EUR/TRY)** — monthly series.
- **Credit Card Sectoral Expenditures** — monthly sector-based spending.

2.2 General Information About the Data

After preprocessing, the data sit in three clean analytical tables.

2.2.1 Exchange Rate Table

Monthly USD/TRY and EUR/TRY rates that capture how far the lira has fallen against major currencies. Both series climbed steadily throughout the sample, with the steepest drops coming after 2021.

2.2.2 Credit Card Expenditure Table

Monthly credit card spending broken down by sector (millions of TRY). We use the total as a rough barometer of how much nominal prices across the economy have risen. As Table 2 shows, e-commerce, supermarkets, food, clothing, and fuel dominate the totals.

2.2.3 Housing Panel Dataset

The heart of the project: Year, Month, Region (NUTS2), House Price Index, and Total Sales. The national Türkiye aggregate row is what we model in the regression.

2.3 Preprocessing

The raw EVDS files arrived as wide Excel sheets with cryptic column codes like `KT2` and `TR62`. We cleaned them in Python (`pandas`): standardising dates, pivoting to long format, translating the codes into readable labels, and rolling up province-level sales to NUTS2 regions to match the HPI geography.

Portions of the preprocessing code were developed with AI assistance; all R analysis, interpretation, and conclusions are the authors' own work.

3 Analysis

3.1 Exploratory Data Analysis

Before looking at relationships, it helps to get a feel for each variable on its own. Table 1 does that.

Table 1: Summary statistics of key variables.

Variable	Min	Mean	Median	Max	SD
House Price Index	7.7	51.1	14.6	204.4	60.8
Monthly Sales (units)	45,218.0	126,140.7	120,437.5	265,223.0	35,718.6
USD/TRY Rate	2.1	12.6	6.1	42.7	12.4
EUR/TRY Rate	2.7	13.9	6.8	49.9	13.7

The HPI's journey from roughly 8 to over 200 gives a sense of just how dramatic the nominal price rise has been. Exchange rates told a similar story; sales, by contrast, bounced around without any clear upward drift — a first hint that rising prices were not being pulled up by rising demand.

Table 2: Top 8 credit card expenditure categories — cumulative total (2014–2025).

Expense Category	Total Spending (Billion TRY)
E-Alışveriş	15,913,640
Market ve Alışveriş Merkezleri	10,409,810
Çeşitli Gıda	3,892,815
Giyim ve Aksesuar	3,889,505
Benzin ve Yakıt İstasyonları	3,542,543
Hizmet Sektörleri	3,486,972
Elektrik-Elektronik Eşya / Bilgisayar	3,308,526
Yemek	3,156,894

We treat total credit card spending as a thermometer for economy-wide inflation rather than a direct measure of housing demand.

3.2 Trend Analysis

3.2.1 House Price Index Over Time

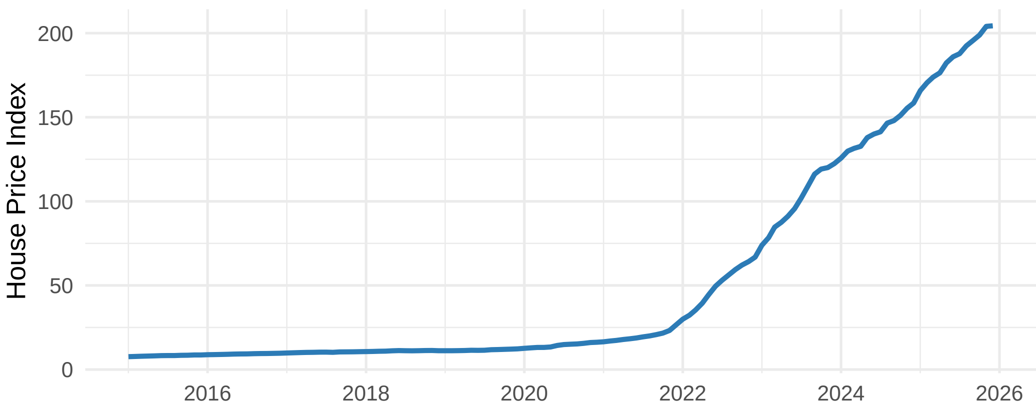

Figure 1: National House Price Index over time (2014–2025).

Things were relatively calm until 2019. Then the index took off (Figure 1), with the sharpest climb running through 2021–2023 — precisely when the lira was losing value fastest and inflation was running hottest. That timing points toward a cost-push story: imported materials got more expensive, expectations shifted, and prices followed.

3.2.2 Year-over-Year Growth Rate Comparison

Comparing raw index levels can be misleading — two series that both trend upward will always look related. Year-over-year growth rates cut through that: if the lira drops sharply in a given year and house prices spike in the same year, that is a much stronger signal.

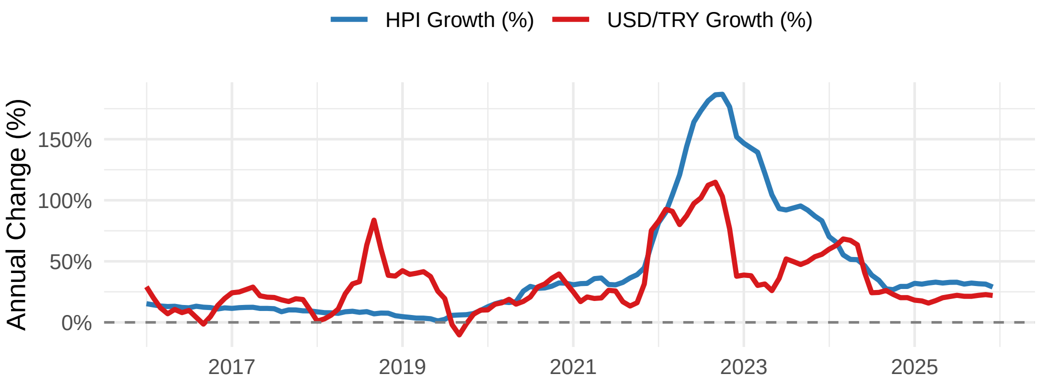

Figure 2: Year-over-year growth rates: House Price Index vs. USD/TRY.

Figure 2 shows that the two series spike together, especially after 2020. That alignment is hard to ignore, though the chart shows co-movement rather than proving one causes the other.

3.2.3 Exchange Rates Over Time

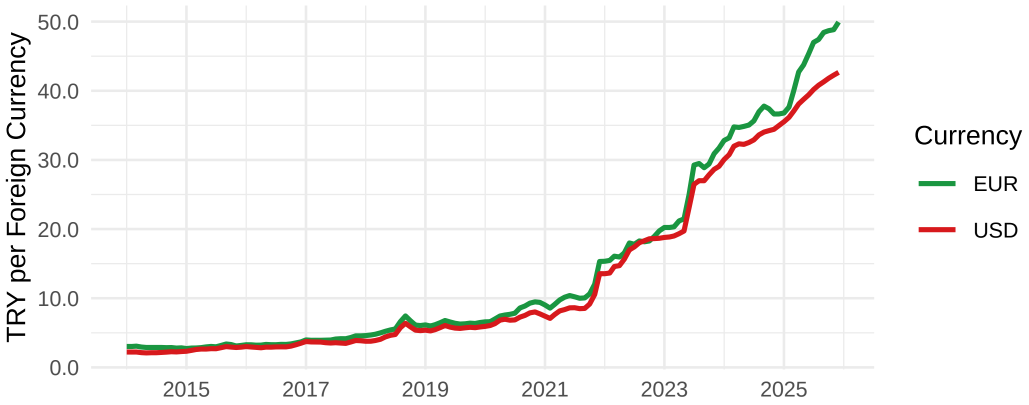

Figure 3: USD/TRY and EUR/TRY exchange rates over the sample period.

3.2.4 Annual Housing Sales

Sales went up and down without ever establishing a clear upward trend (Figure 4). Prices were rising, but people were not buying more homes — if anything, the opposite. That is not what demand-driven inflation looks like.

3.3 Correlation Analysis

3.3.1 Correlation Heatmap

Figure 5 gives a bird's-eye view of how all the variables relate to each other at once.

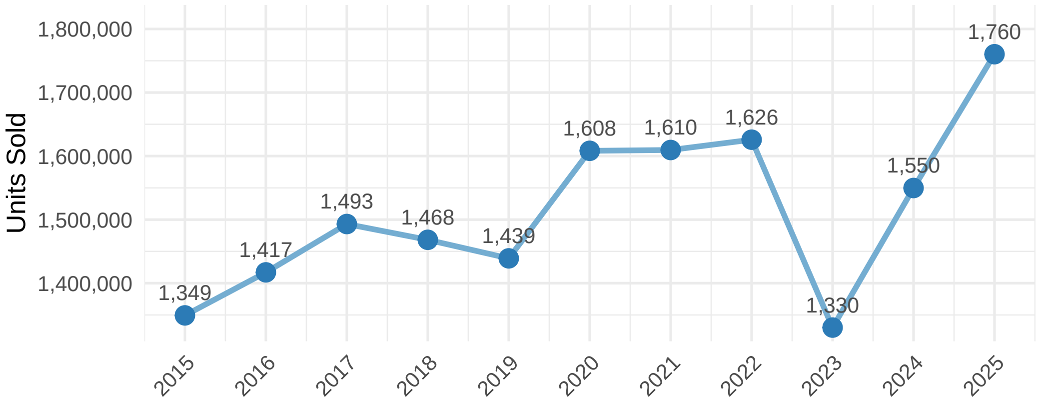

Figure 4: Annual housing sales for Türkiye. Labels show thousands of units sold.

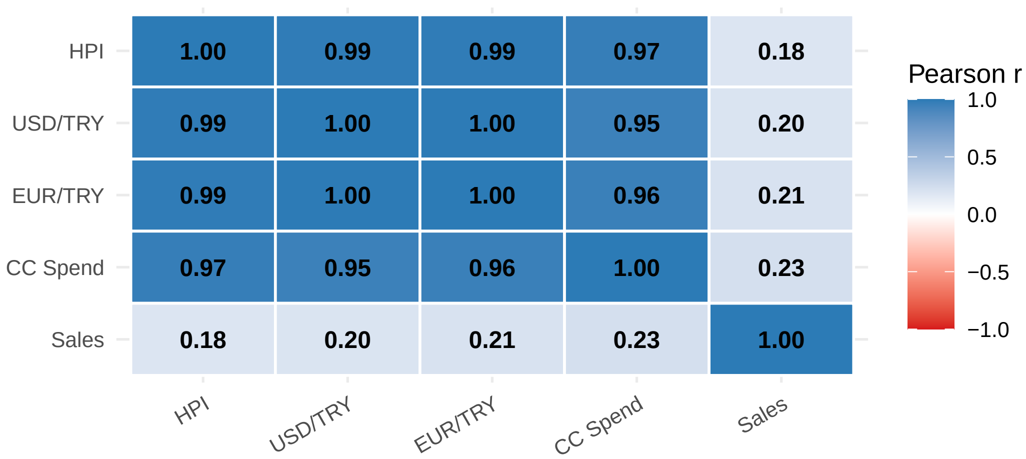

Figure 5: Pearson correlation heatmap of key variables.

Table 3: Pearson correlations with the House Price Index.

Variable Pair	Pearson r
HPI vs USD/TRY	0.990
HPI vs EUR/TRY	0.990
HPI vs CC Spend	0.973
HPI vs Sales	0.184

The numbers are striking: the HPI moves almost in lockstep with exchange rates ($r \approx 0.99$) and credit card spending ($r \approx 0.97$), while its link to sales is barely there ($r \approx 0.18$, Table 3). It is worth keeping in mind that all nominal series trend upward, which can inflate correlation coefficients; the YoY analysis and regression models are designed to address exactly that.

3.3.2 Scatter Plot: log(HPI) vs. log(USD/TRY)

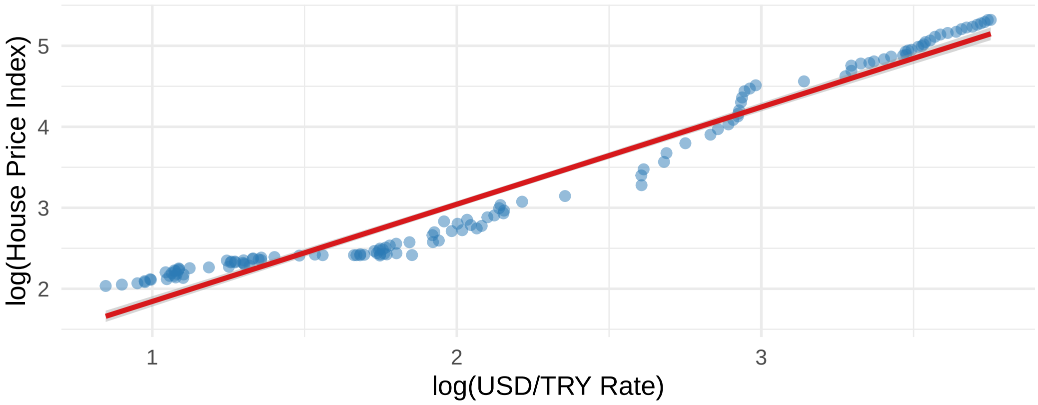

Figure 6: Scatter plot of log(HPI) against log(USD/TRY). Each point is one monthly observation.

Figure 6 makes the point more directly: even when you strip out the time dimension and just plot one month against another, the log-linear relationship holds tight across the entire range of exchange rate values.

3.4 Regional Comparison

3.4.1 Regional HPI Trajectories

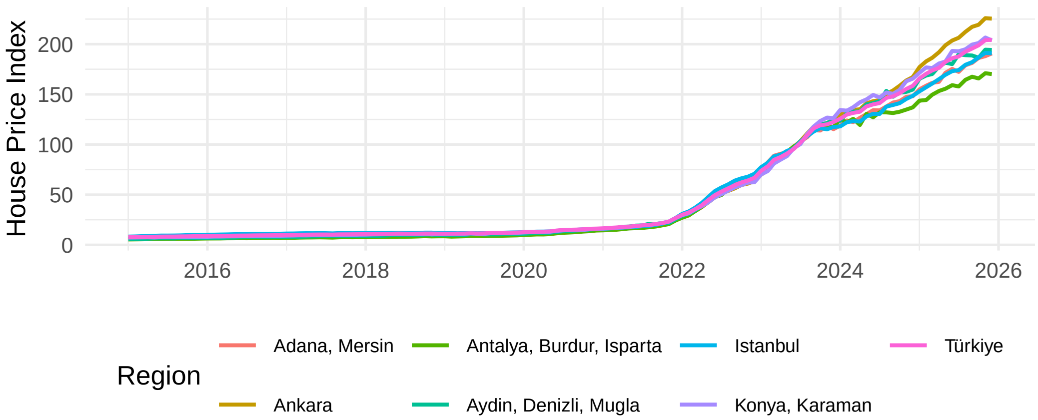

Figure 7: House Price Index trajectories for selected regions (2014–2025).

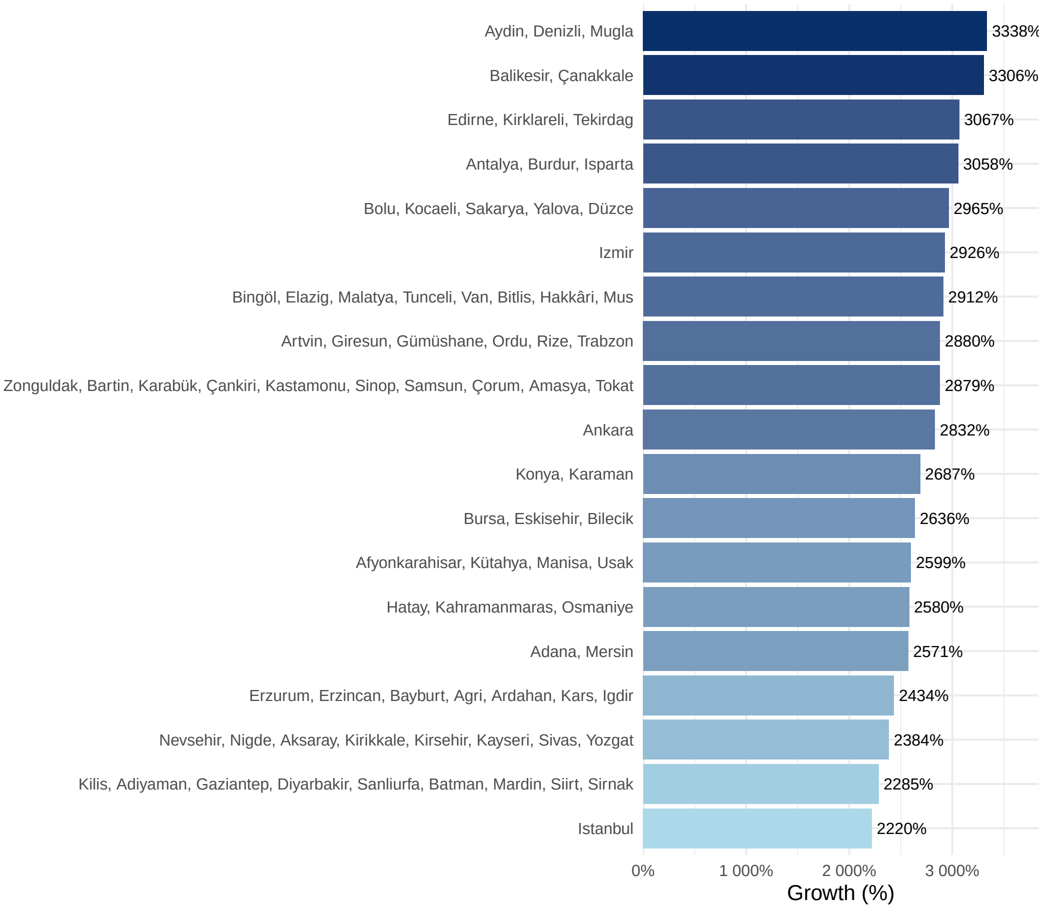

Figure 8: Total HPI percentage growth by NUTS2 region (first to latest observation).

Table 4: HPI growth by NUTS2 region (first to latest observation).

Region	First HPI	Latest HPI	Growth (%)
Aydın, Denizli, Muğla	5.7	194.2	3337.5
Balıkesir, Çanakkale	6.2	211.9	3306.1
Edirne, Kırklareli, Tekirdağ	6.2	196.1	3067.2
Antalya, Burdur, Isparta	5.4	170.2	3057.5
Bolu, Kocaeli, Sakarya, Yalova, Düzce	6.9	210.9	2964.8
İzmir	6.6	198.8	2926.5
Bingöl, Elazığ, Malatya, Tunceli, Van, Bitlis, Hakkâri, Muş	8.2	245.8	2912.4
Artvin, Giresun, Gümüşhane, Ordu, Rize, Trabzon	7.4	219.7	2880.3
Zonguldak, Bartın, Karabük, Çankırı, Kastamonu, Sinop, Samsun, Çorum, Amasya, Tokat	7.3	217.8	2879.1
Ankara	7.7	225.5	2831.9
Konya, Karaman	7.3	203.8	2687.3
Bursa, Eskişehir, Bilecik	7.4	203.0	2636.4
Afyonkarahisar, Kütahya, Manisa, Uşak	8.5	228.9	2599.2
Hatay, Kahramanmaraş, Osmaniye	8.1	216.3	2580.2
Adana, Mersin	7.1	190.7	2571.4
Türkiye	7.7	204.4	2571.4
Erzurum, Erzincan, Bayburt, Ağrı, Ardahan, Kars, Iğdır	9.8	247.8	2433.7
Nevşehir, Niğde, Aksaray, Kırıkkale, Kırşehir, Kayseri, Sivas, Yozgat	8.8	219.1	2384.5
Kilis, Adıyaman, Gaziantep, Diyarbakır, Şanlıurfa, Batman, Mardin, Siirt, Şırnak	8.8	210.4	2285.4
İstanbul	8.2	190.9	2219.8

3.4.2 Regional Growth Ranking

The Aegean and Marmara coastal areas saw the biggest percentage gains (Table 4), likely pushed up by internal migration, tourism, and second-home buyers. İstanbul sits at the top in absolute terms. But perhaps the most striking finding is that every single region exceeds 2,000% nominal appreciation — this was not a localised phenomenon.

3.5 Model Fitting

To put numbers on these relationships, we fit log-log regression models, where each coefficient reads approximately as an elasticity. Because all series trend upward together, we treat the results as measuring association rather than claiming causation.

3.5.1 Baseline Model — Exchange Rate Only

$$\log(\mathrm{HPI}_t) = \beta_0 + \beta_1 \log(\mathrm{USD/TRY}_t) + \varepsilon_t$$

Table 5: Baseline model coefficients. $R^2 = 0.959$.

	Term	Estimate	Std. Error	t value	p value
(Intercept)	Intercept	0.644	0.052	12.49	0.0000
log_USD	log(USD/TRY)	1.200	0.022	55.03	0.0000

A single variable — the USD/TRY rate — accounts for roughly 96% of the movement in log(HPI). The elasticity of $\hat{\beta}_1 \approx 1.20$ means that every 1% the lira loses against the dollar is associated with house prices rising by about 1.20%.

3.5.2 Full Model — Exchange Rate, Sales, and Credit Card Expenditure

$$\log(\mathrm{HPI}_t) = \beta_0 + \beta_1 \log(\mathrm{USD}_t) + \beta_2 \log(\mathrm{Sales}_t) + \beta_3 \log(\mathrm{CC}_t) + \varepsilon_t$$

Table 6: Full model coefficients. $R^2 = 0.986$, Adjusted $R^2 = 0.985$.

	Term	Estimate	Std. Error	t value	p value	Sig
(Intercept)	Intercept	-8.956	0.820	-10.92	0.0000	***
log_USD	log(USD/TRY)	0.256	0.062	4.12	0.0001	***
log_Sales	log(Sales)	-0.120	0.045	-2.68	0.0084	**
log_CC	log(CC Spend)	0.679	0.044	15.55	0.0000	***

Adding sales and credit card spending pushes adjusted R^2 to around 0.985. The exchange rate is still the engine of the model. The negative sign on sales fits the story told by Figure 4: as prices went up, buyers pulled back. The credit card coefficient is picking up broad inflation rather than anything specific to housing.

3.5.3 Actual vs. Fitted Values

The model follows the actual HPI surprisingly closely (Figure 9), including the sharp kink after 2021. The main place it struggles is 2022, when the lira's slide was at its most extreme — a stretch that probably demands a more flexible model than a simple log-linear form.

3.5.4 Residual Plot

For most of the fitted range the residuals sit comfortably near zero (Figure 10). The slight bow at the upper end points back to 2022–2023, confirming that is where the model's assumptions are most stretched.

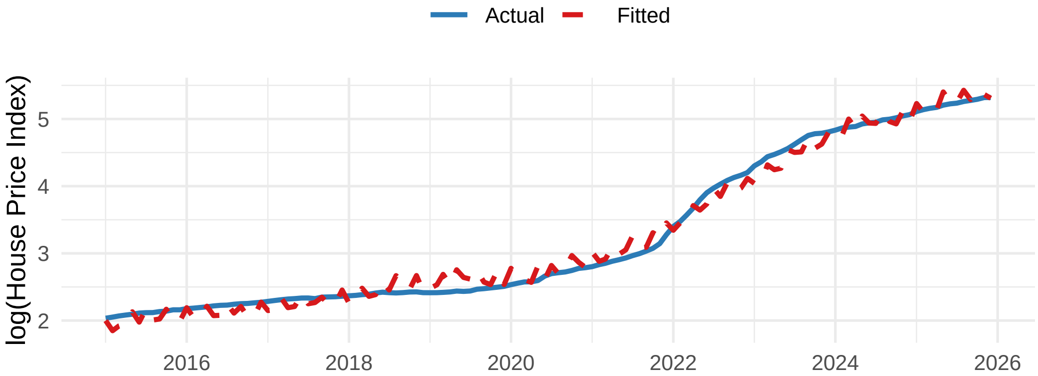

Figure 9: Actual and fitted log(HPI) — full model.

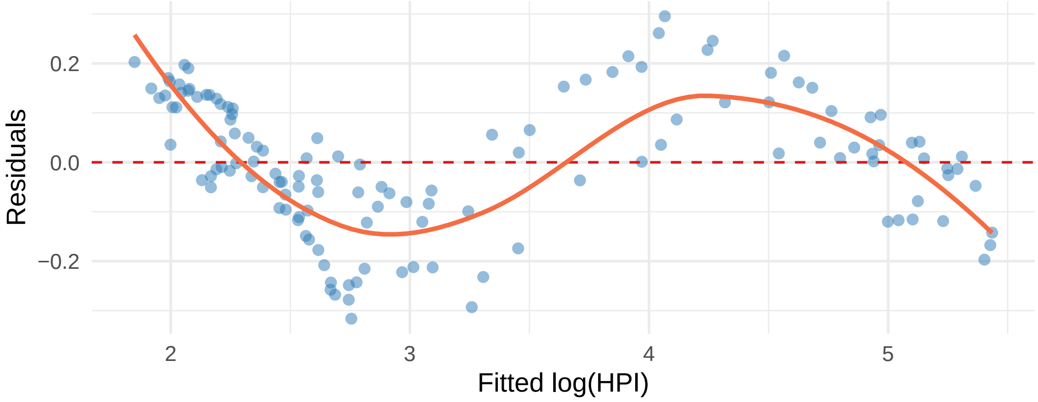

Figure 10: Residuals vs. fitted values — full model. The LOESS smoother (orange) reveals mild non-linearity at high fitted values.

4 Results and Key Takeaways

1. **House prices surged after 2020.** The HPI more than doubled during 2021–2023 — a rise that has no precedent in the modern Turkish data.
2. **The lira's fall is the single strongest correlate** ($r \approx 0.99$, baseline elasticity ~1.20). Lose a percent on the currency, gain a percent on the house.
3. **More transactions did not drive the price rise.** Sales bounced around without a trend while prices climbed, which looks much more like demand being squeezed out than being stoked.
4. **Credit card spending reflects economy-wide inflation**, not housing-specific demand — useful as a control variable, not a cause.
5. **The surge is national, not just metropolitan.** Every region crossed 2,000% nominal appreciation; coastal areas led on percentage growth, İstanbul on absolute level.
6. **Strong correlations do not mean proven causality.** All variables trend upward together, so the results should be read as consistent with — not definitive proof of — the exchange-rate story.
7. **The next step is causal identification.** Real (inflation-adjusted) prices, mortgage data, and time-series cointegration methods would let future work make stronger claims.

5 Conclusion

The data tell a fairly consistent story: when the lira falls, house prices rise, and the relationship is tight enough that a single exchange-rate variable explains roughly 96% of the movement in log(HPI). Bringing in sales and credit card spending gets the full model to ~99%. If that story is right, then policy responses aimed only at the housing market — new zoning rules, transaction taxes, caps on purchases — are probably fighting the symptom rather than the cause. Stabilising the currency and bringing down broad inflation seems more to the point. The main caveat is that all our variables are nominal and move upward together, so the association, however tight, falls short of proof. Richer data and time-series cointegration methods are the natural next step.

References

Central Bank of the Republic of Türkiye (TCMB). (2024). *Electronic Data Delivery System (EVDS)*. Retrieved from https://evds2.tcmb.gov.tr/